

Navigation2 Multi-Point Navigation

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1. Course Content

- Run the sample program to learn how to manually mark multiple waypoints for navigation and obstacle avoidance.

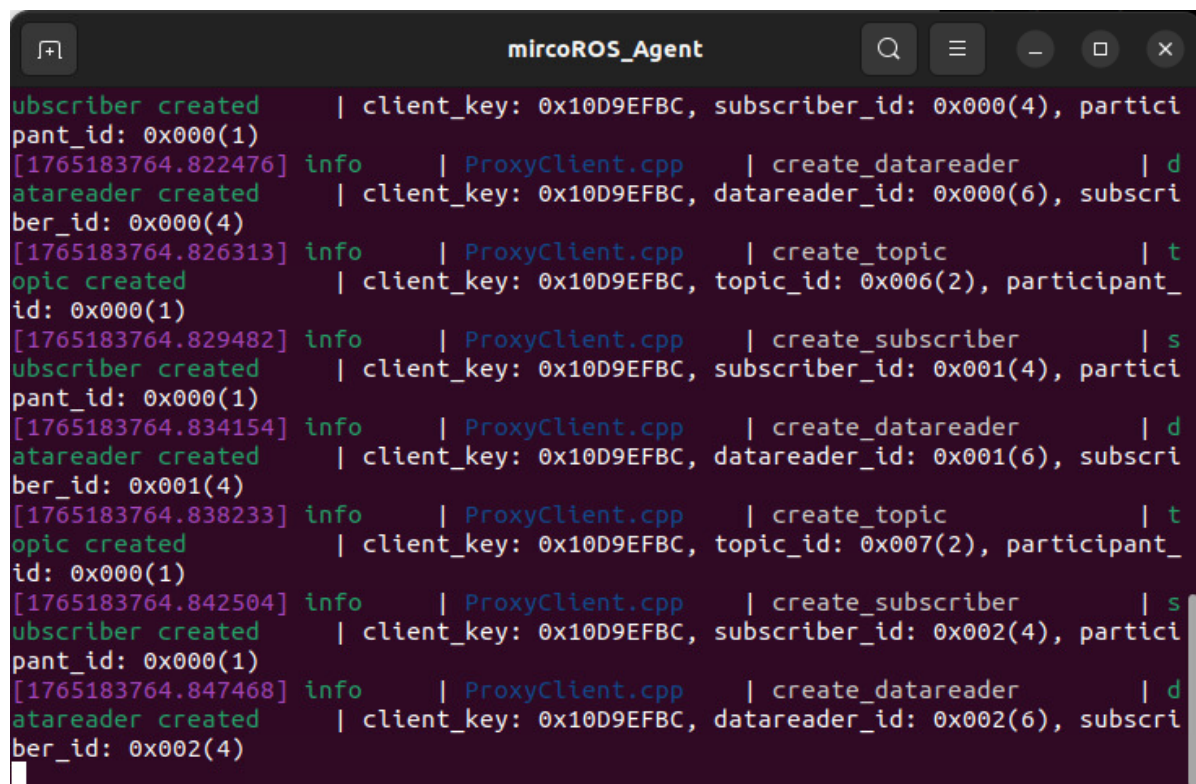
2. Start the Agent

Note: If it is already running, you do not need to start it again.

Enter the command in the vehicle's terminal:

```
start_agent
```

The terminal will print the following information, indicating a successful connection:



```
mircoROS_Agent
[1765183764.822476] info | ProxyClient.cpp | create_datareader | d
datareader created | client_key: 0x10D9EFBC, datareader_id: 0x000(6), subscri
ber_id: 0x000(4)
[1765183764.826313] info | ProxyClient.cpp | create_topic | t
topic created | client_key: 0x10D9EFBC, topic_id: 0x006(2), participant_
id: 0x000(1)
[1765183764.829482] info | ProxyClient.cpp | create_subscriber | s
subscriber created | client_key: 0x10D9EFBC, subscriber_id: 0x001(4), partici
pant_id: 0x000(1)
[1765183764.834154] info | ProxyClient.cpp | create_datareader | d
datareader created | client_key: 0x10D9EFBC, datareader_id: 0x001(6), subscri
ber_id: 0x001(4)
[1765183764.838233] info | ProxyClient.cpp | create_topic | t
topic created | client_key: 0x10D9EFBC, topic_id: 0x007(2), participant_
id: 0x000(1)
[1765183764.842504] info | ProxyClient.cpp | create_subscriber | s
subscriber created | client_key: 0x10D9EFBC, subscriber_id: 0x002(4), partici
pant_id: 0x000(1)
[1765183764.847468] info | ProxyClient.cpp | create_datareader | d
datareader created | client_key: 0x10D9EFBC, datareader_id: 0x002(6), subscri
ber_id: 0x002(4)
```

3. Run the Example

3.1 Multi-Point Navigation

- Start the navigation function:

For Raspberry Pi 5 users, if the ROS container is not started, you need to start it first,

```
bringup_m3
```

Open a terminal inside the container,

```
exec_m3
```

(ORIN board)

```
ros2 launch m3_bringup nav2.launch.py
```

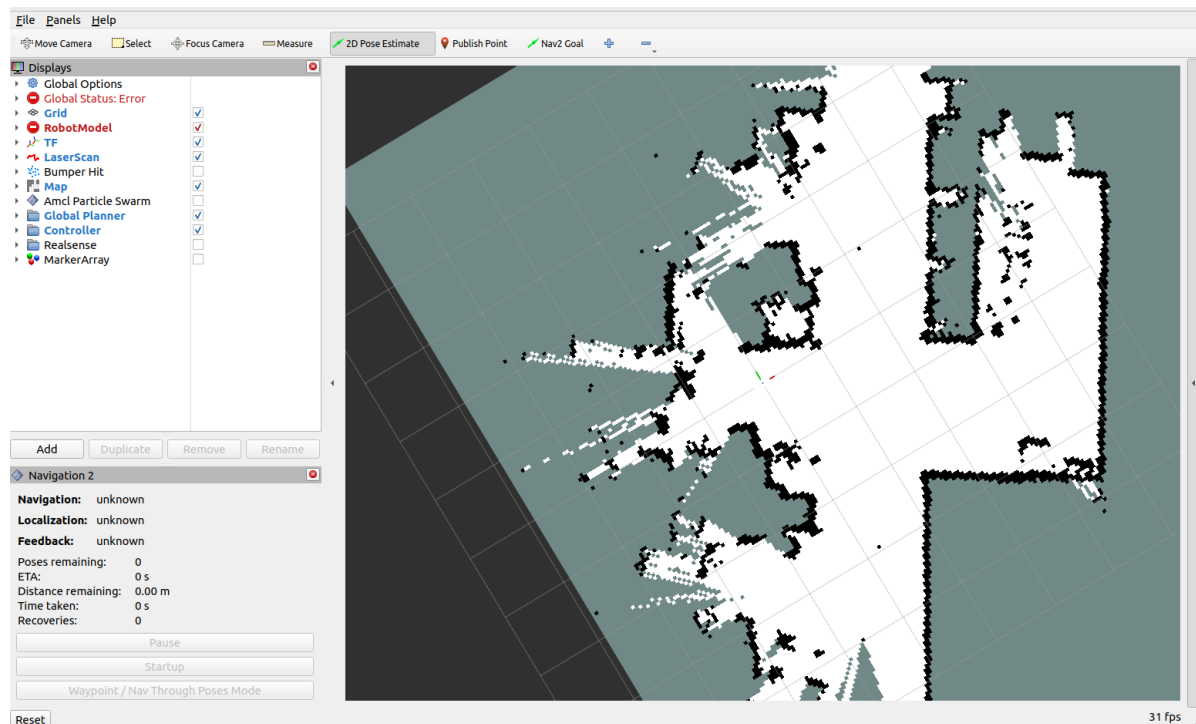
(RDK & PI5 boards) Run in separate steps, with visualization launched on the accompanying virtual machine `Rosmaster-M3-Ubuntu22.04`,

On the virtual machine

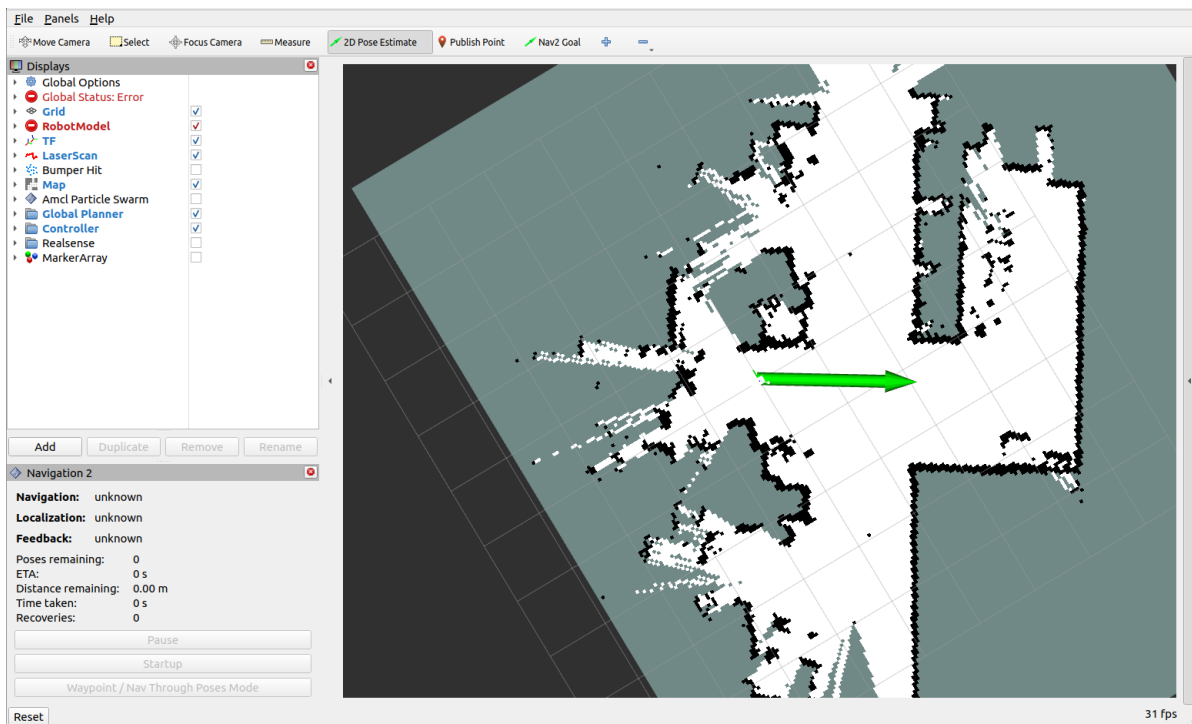
```
rviz2 -d /home/yahboom/yahboomM3_ws/user_ws/src/m3_bringup/rviz/nav2.rviz
```

On the host machine

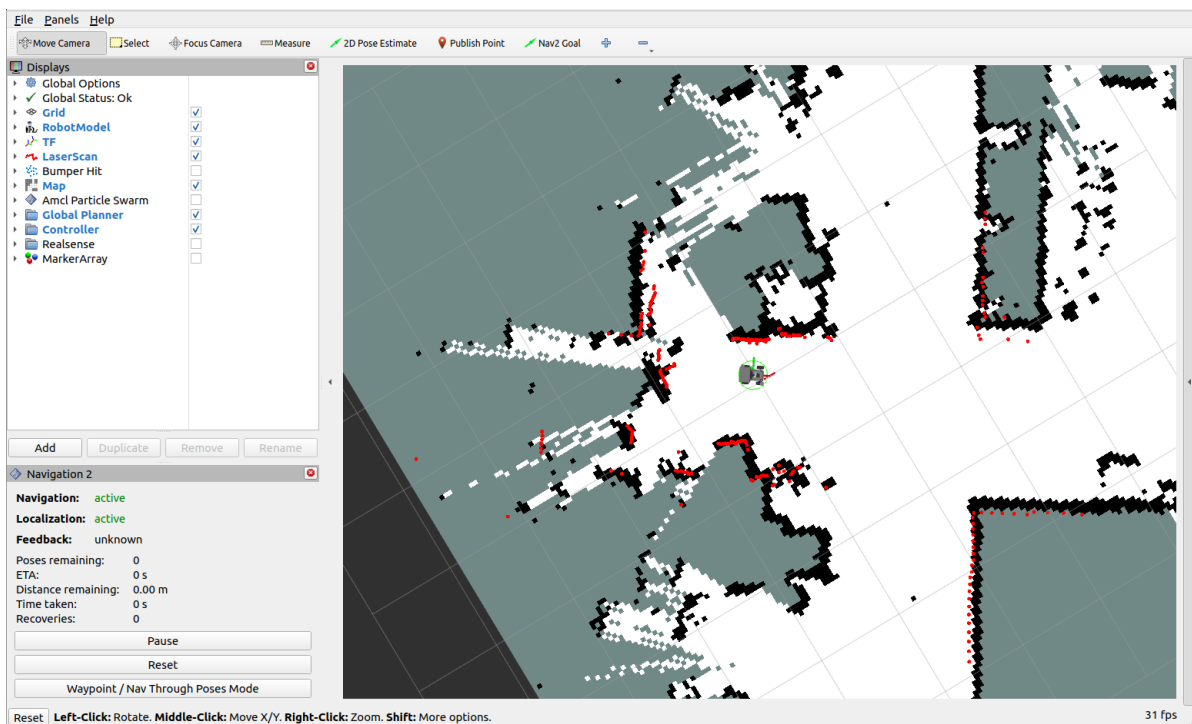
```
ros2 launch m3_bringup nav2.launch.py gui:=False
```



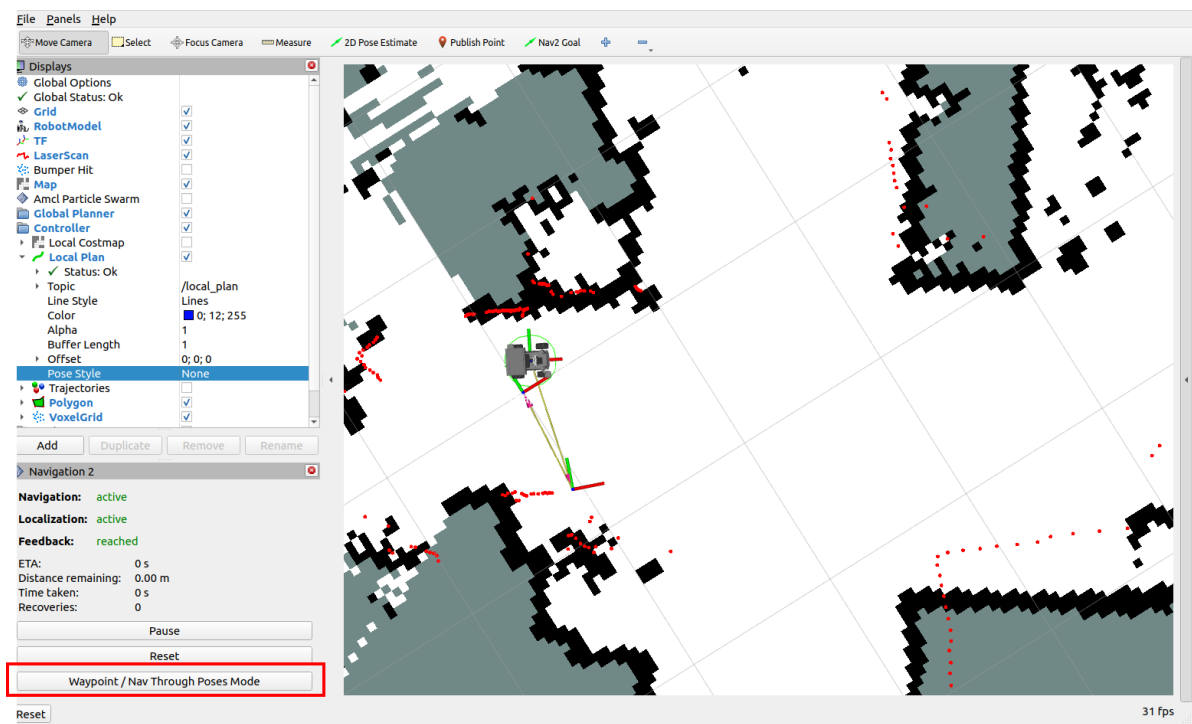
- At this point, you can see the map loaded. Then, click on "2D Pose Estimate" to set the initial pose of the car. Based on the car's actual position in the environment, click and drag with the mouse in RViz. The car model will move to the position we set. As shown in the figure below, if the Lidar scan area roughly coincides with the actual obstacles, the pose is accurate.



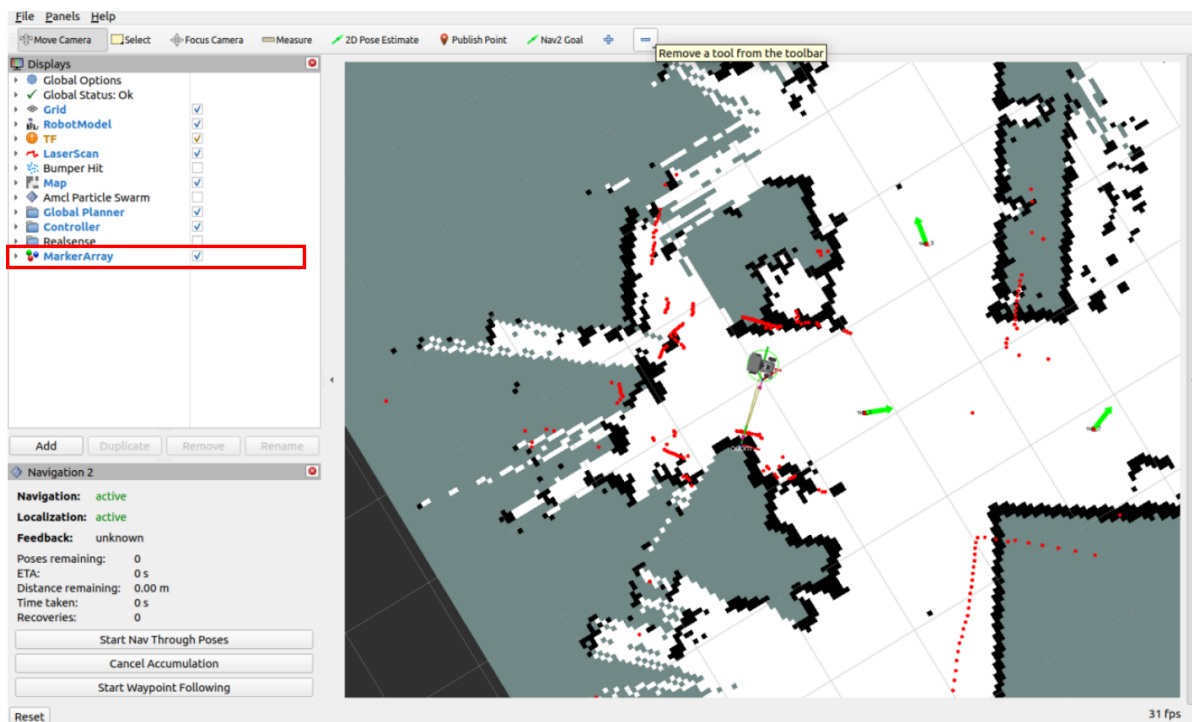
- After the pose is initialized, the robot model and the red 2D Lidar point cloud will appear in the RViz interface.



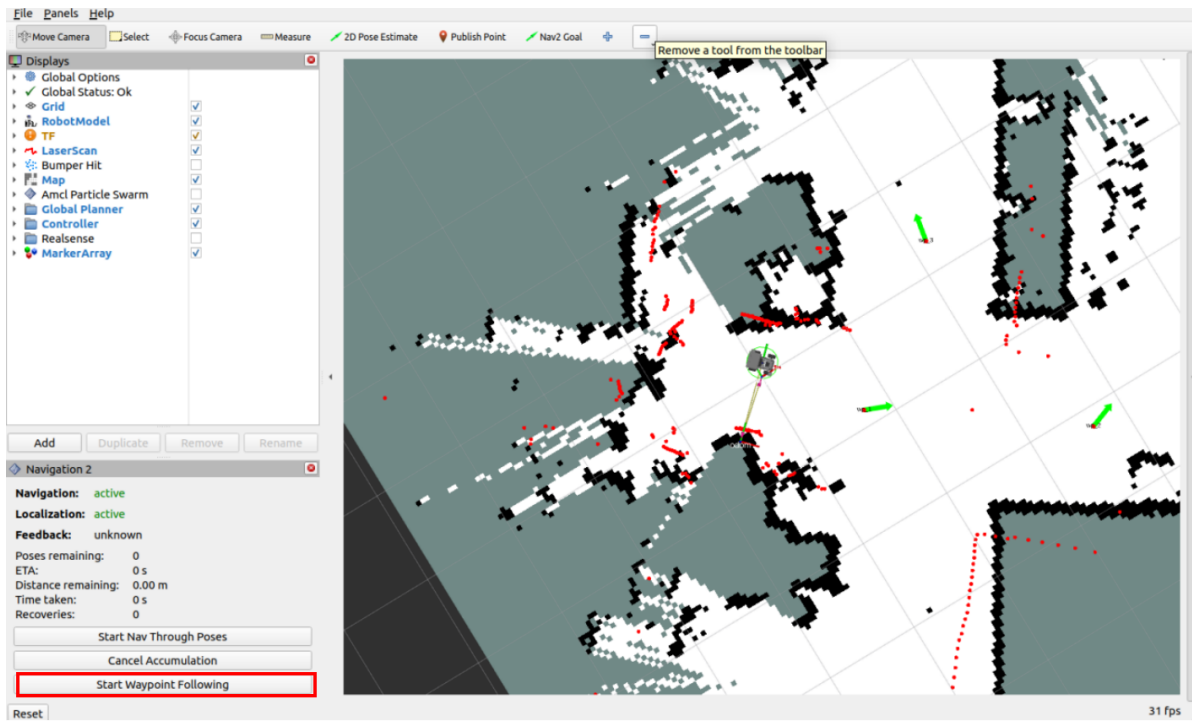
- Click on **[Waypoint/Nav Through PoseMode]** in the lower-left corner to enter multi-point navigation mode.



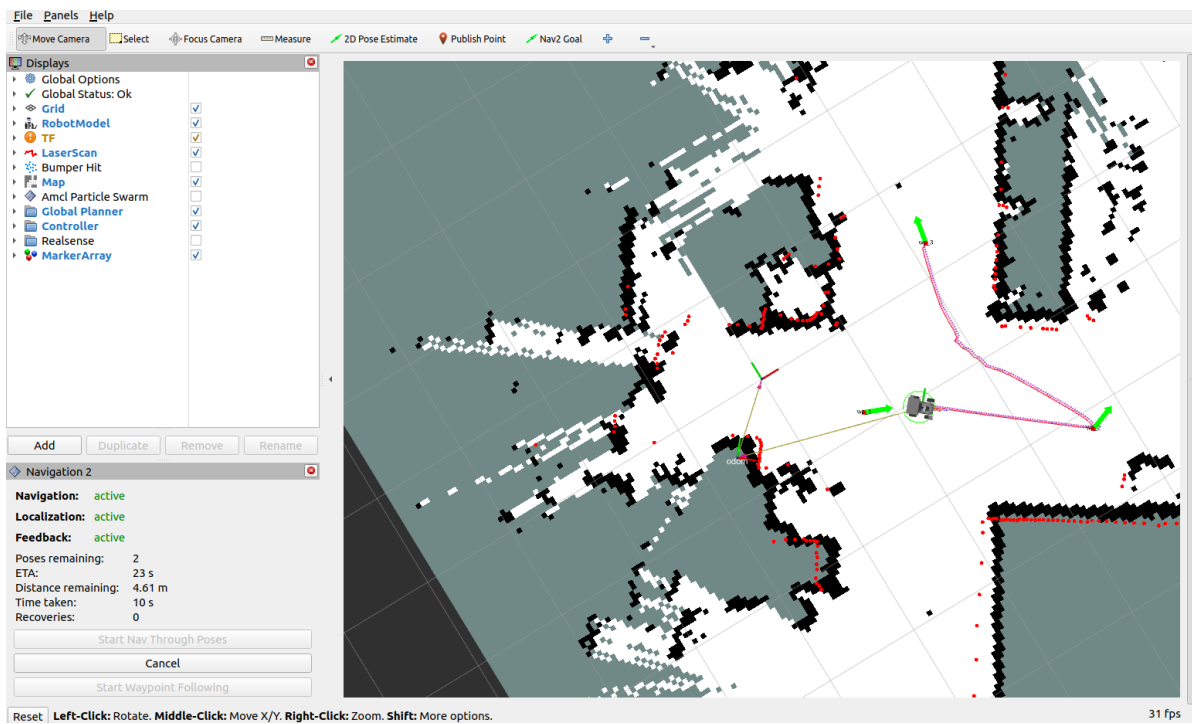
- Check the **MarkerArray** in the left options panel to display the waypoints, then click **Nav2 Goal** and use the mouse to mark multiple target points on the map.



Click **Start Waypoint Following** in the lower-left corner to begin multi-point navigation.



- The robot car will navigate sequentially to the marked points.



4. Principle Analysis

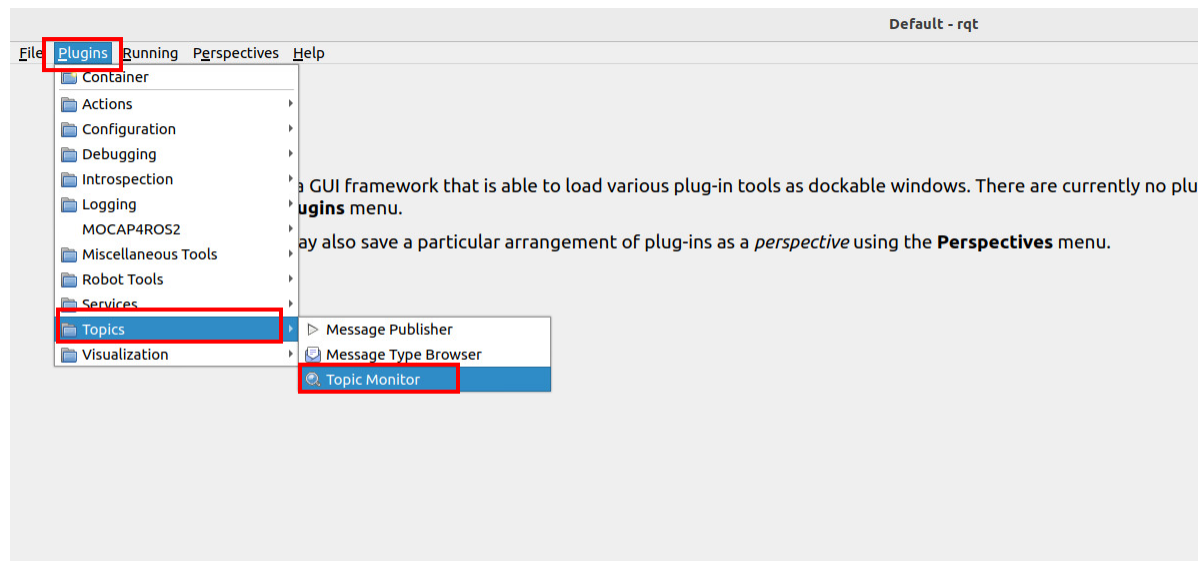
4.1 View Waypoint Data

After the user opens **[Waypoint/Nav Through PoseMode]** and enters multi-point navigation mode, the marked point information will be published to the `/waypoints` topic (the RViz waypoint navigation plugin will add extra waypoints between the target waypoints to smooth the path). We can view this **waypoint data** through the RQT interface.

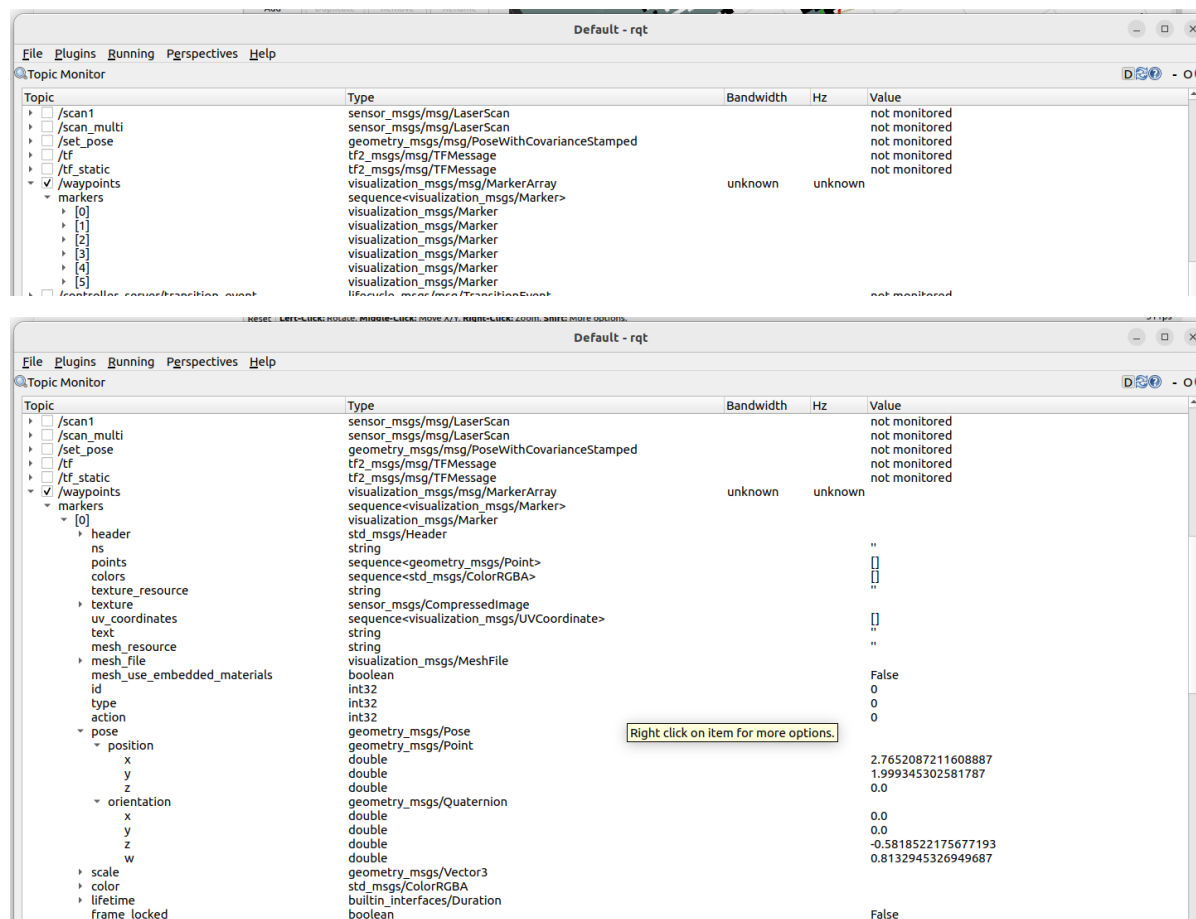
Launch command in the virtual machine terminal:

```
rqt
```

Click Plugins -> Topic -> Topic Monitor



- In the RQT interface, you can see the `/waypoints` topic. After checking it, you can observe the data on the topic (you need to check the topic first, then publish the waypoints in the RViz interface).
- The waypoints we manually mark in RViz will be published to this topic. You can expand one of the waypoints to view its data. Taking [0] as an example, `pose` contains the coordinate data.



4.2 Data Sending and Execution

- After setting the waypoint coordinates, when you click **Start Waypoint Following**, the RViz plugin will package the sequence of waypoint coordinates into a `FollowWaypoints` action request and send it to the `/follow_waypoint` action server to execute all the waypoints sequentially.
- View the node relationship graph:

```
ros2 run rqt_graph rqt_graph
```

- In the node relationship graph, you can see the `/follow_waypoint` action server, which receives the waypoint sequence and navigates through them in order.

